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TE COMP - Div D Batch A

SE LAB

## **Experiment 4: Sequence and Collaboration Diagrams**

### **Use Case Diagram:**

#### **Use Cases selected to make the diagrams**

| <b>Use Case</b>   | <b>Description</b>                                                                                                                                                                                                                                                                                                                                                          |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Join a Group      | This use case allows a registered user to become a member of an existing group. The user selects a group of interest, and the system verifies eligibility. If the group is public, the user is added immediately; if it is private, the system forwards a request to the admin for approval. Once joined, the user can access group discussions, resources, and activities. |
| Search for Skills | This use case enables users to discover groups or topics relevant to their learning interests. The user enters keywords or skill names into the search bar, and the system retrieves matching groups or topics. If results are found, they are displayed to the user for exploration and potential joining; if not, the system suggests alternatives.                       |

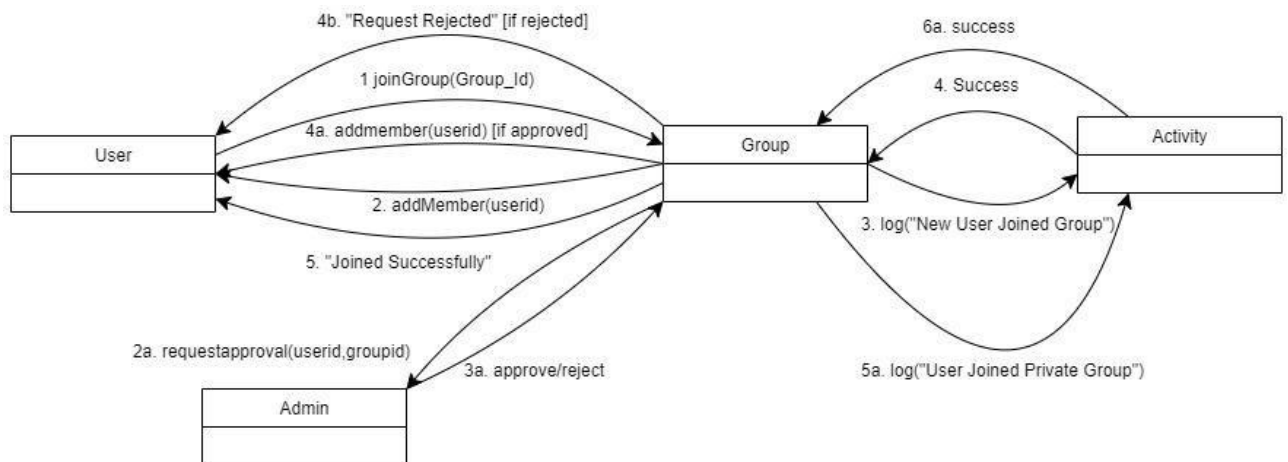
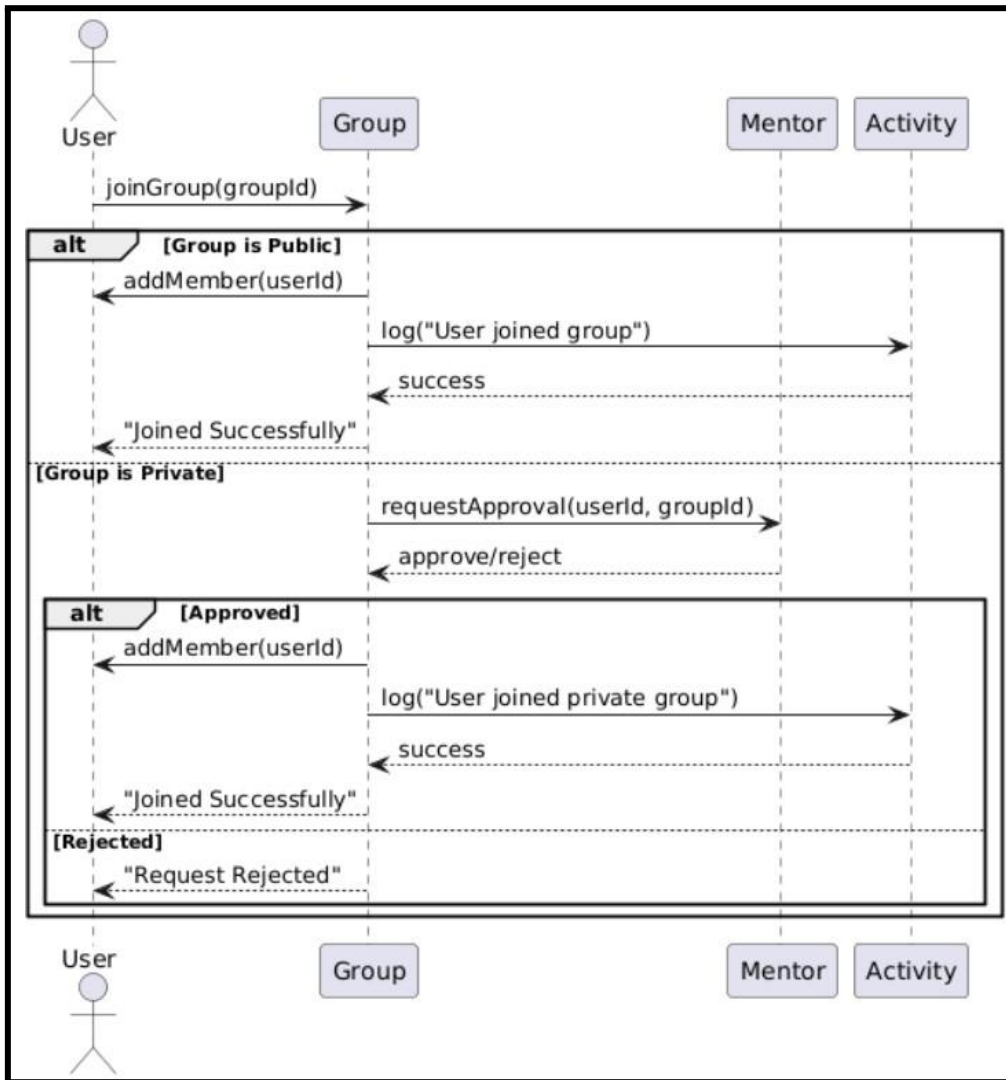
### **Sequence Diagram:**

A Sequence Diagram is a type of interaction diagram in UML that shows how objects interact over time. It emphasizes the time order of messages between objects. The diagram is read top to bottom, where lifelines represent participating objects and arrows represent messages or method calls.

### **Collaboration Diagram:**

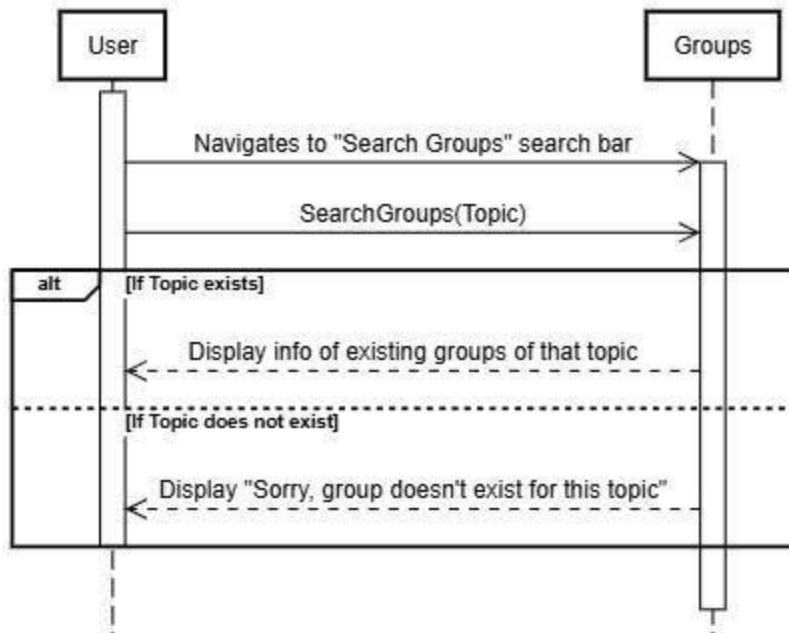
A Collaboration Diagram (UML 1.x) or Communication Diagram (UML 2.x) shows how objects interact but focuses on the structural relationships between them instead of time sequence. Objects are connected by links, and messages are represented by numbered arrows indicating the order of communication.

## Diagrams for Use Case : Join a Group

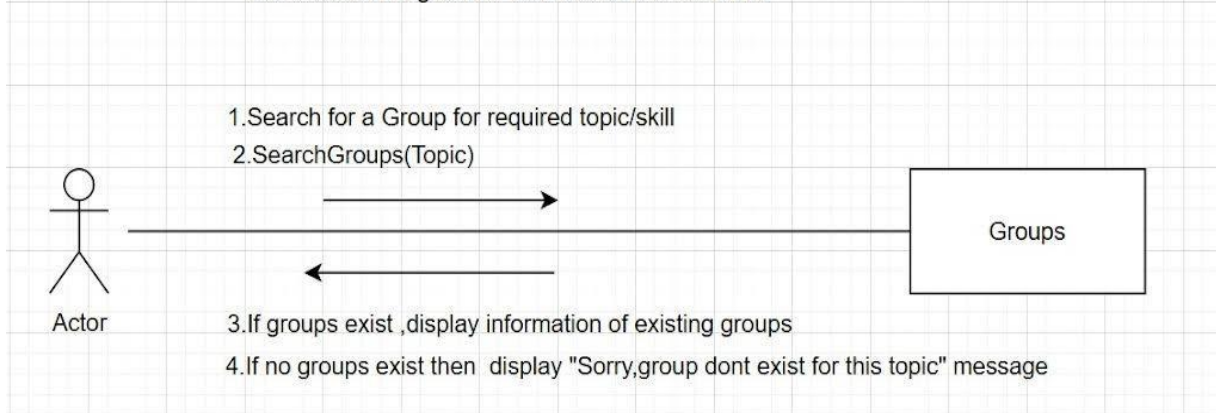


## Diagrams for Use Case : Search Skills

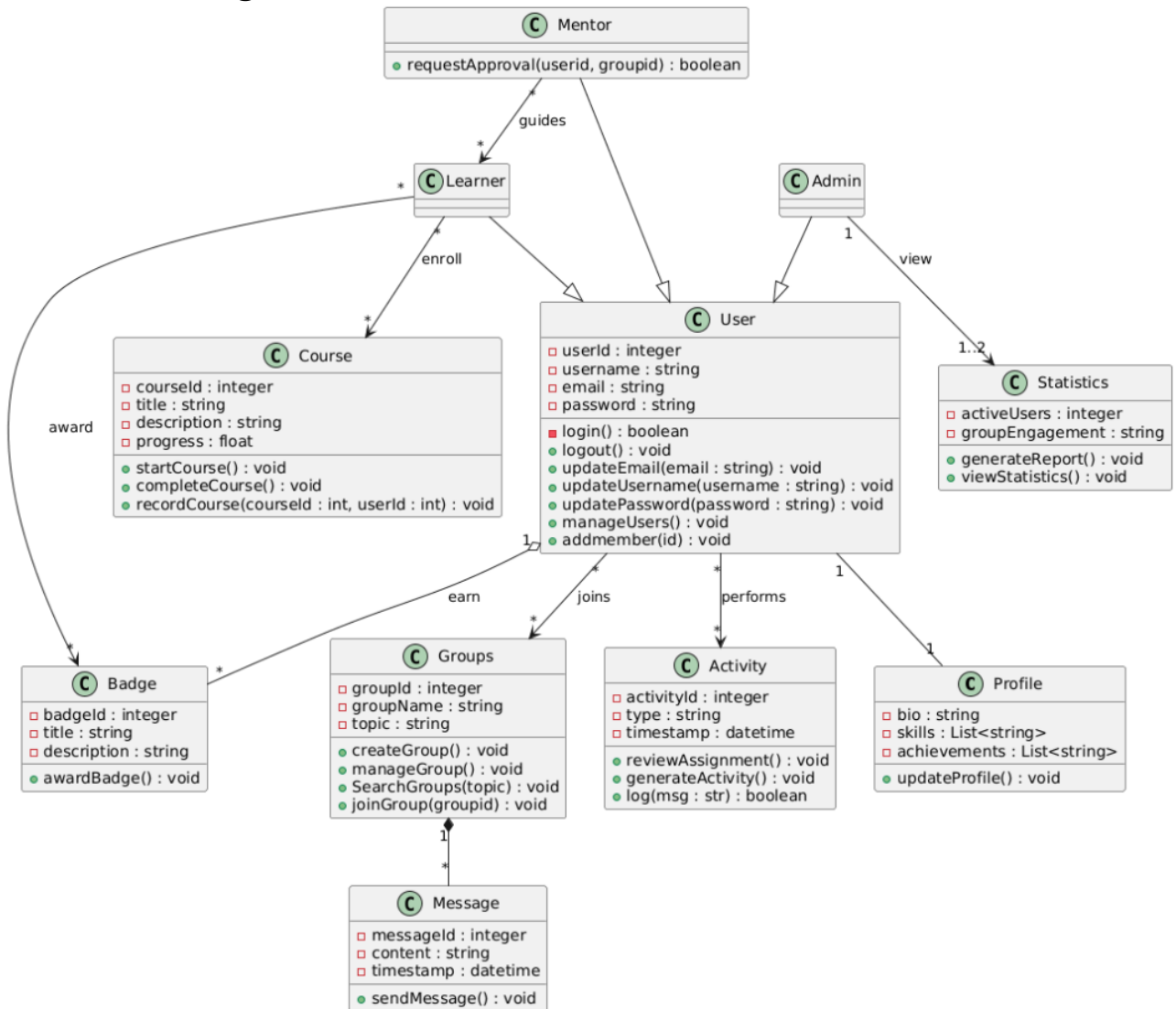
Sequence Diagram: Search Groups



Collabration Diagram for Use Case : Search Skills



## Corrected Class Diagram:



## Conclusion:

In this experiment, we modeled the dynamic behavior of the Collaborative Skill-Learning Platform using Sequence and Collaboration diagrams. The Sequence Diagram clearly depicted the order of interactions among classes (User, Group, Admin, Activity) for use cases emphasizing the flow of messages over time. The Collaboration Diagram, on the other hand, represented the same use case by highlighting the relationships among objects and numbering their communications. Together, these diagrams provide a complete view of both control flow and structural interactions, making system behavior easier to understand, validate, and communicate to developers and stakeholders.